Q1.

The different parts of the electromagnetic spectrum are shown below.

Gamma rays	X-rays	Ultraviolet	Visible light	Infrared	Microwaves	Radio waves

(a) Name a part of the electromagnetic spectrum with:

(i) a longer wavelength than microwaves: _____

(1)

(ii) greater energy than X-rays: _____

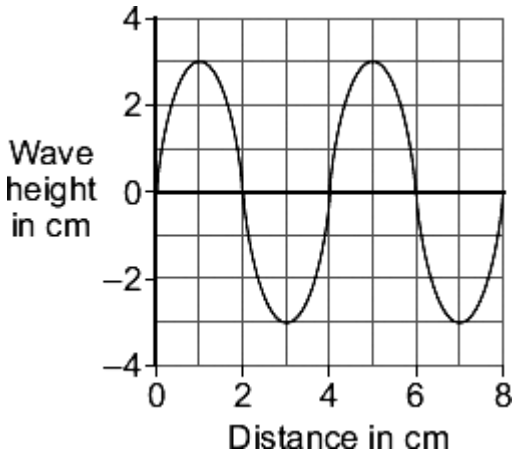
(1)

(iii) a higher frequency than ultraviolet: _____

(1)

(b) The properties of water waves can be measured easily in a school lab.

The diagram shows information about waves.



(i) How many complete waves are shown in the diagram?

_____

(1)

(ii) What is the wavelength of each wave in the diagram?

_____ cm

(1)

(iii) What is the amplitude of the waves?

_____ cm

(1)

(iv) Complete the sentence below.

The oscillations of the waves in the diagram are perpendicular to the direction of energy transfer. They are called _____ waves.

(1)

(Total 7 marks)

Q1.

(a) (i) radio (waves)

1

(ii) gamma (rays)

accept γ

do ***not*** *accept* α

1

(iii) X-rays
or
gamma (rays)

accept γ

do ***not*** *accept* α

1

(b) (i) 2

1

(ii) 4(cm)

1

(iii) 3(cm)

ignore + or – in front of 3(cm)

1

(iv) transverse

1

[7]